ISBN 978-1-5386-4340-2

Proceedings
of the 2018 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering
(ElConRus)

January 29 - February 01, 2018

St. Petersburg and Moscow
Russia
2018

Contents

Part 1.
Conference venue at St. Petersburg Electrotechnical University “LETI”, Saint Petersburg

Section 1. Communication. Networking & Broadcasting

Section 2. Components, Circuits, Devices & Systems

Section 4. Engineered Materials, Dielectrics & Plasmas

Section 5. Fields, Waves & Electromagnetics

Section 6. Photonics & Electro-Optics

Section 7. Power, Energy & Industry Applications

Section 8. Robotics & Control Systems

Section 9. Signal Processing & Analysis

Section 10. Biomedical and Environmental Engineering

Section 11. Industrial Management, Management of Innovations

Part 2.
Conference venue at the National Research University of Electronic Technology “MIET”, Moscow

Section 1. Communication, Networking & Broadcasting

Section 2. Computing & Processing, Hardware

Section 3. Computing & Processing, Software

Section 5. Fields, Waves & Electromagnetics

Section 6. Electrical Engineering, Automation & Control

Section 7. Digital Signal, Image & Video Processing & Analysis

Section 8. Biomedical and Environmental Engineering

Section 9. Nanoelectronics

The Performance Measurement Test on Rain Gauge of Tipping Bucket due to Controlling of the Water Flow Rate

Jalu.A. Prakosa, Sensus Wijonarko, Dadang Rustandi
Research Center for Metrology – Indonesian Institute of Sciences (RCM – LIPI)
Tangerang Selatan, Indonesia
jalu001@lipi.go.id

***Abstract*—The purpose of this research was to test the measurement performance especially the measurement repeatability of tipping bucket rain gauge by varying the water intensities. Water intensities were obtained by converting water (volume) flow rates that enter to the section or mouth area of the rain gauge. The study was conducted using an experiment method. The result showed that by changing the water flow rate between 24 – 100 ml/min three times, the water volume in the tipping bucket were (18.93±0.60), (17.89±0.92) and (17.80±0.60) ml/tip respectively.**

Keywords— water flow rate; intensity; repeatability; tipping bucket rain gauge; uncertainty

I. INTRODUCTION

Tipping bucket is the most widely used rain gauge (hyetometer, ombrometer, pluviometer, regenmeter or udometer[1]) particularly in meteorology institutes[2, 3] and perhaps in the world because tipping bucket rain gauge has ever been intercompared[4]. This gage is also one of the most preferred rain gauge in Indonesia[5]. These may be caused by its simplicity in working principle, reliability, and easiness to connect to a further processing system. To ensure the measurement quality of tipping bucket rain gauge, its performance should be tested first[6].

In the real world, rainfall (rain amount) has variation in its intensity. Rains are called as for instance, light rain for the rain intensity < 2.5 mm per hour, moderate rain between 2.5 mm - 7.6 mm per hour and heavy rain > 7.6 mm per hour[7]. Hence, it is reasonable that the tipping bucket testing should be conducted as a function the rain intensity. In this study, this was carried out by controlling the water flow rate before it entered to the tipping bucket. The water flow rate was converted to rain intensity by connecting both variables with the mouth of tipping bucket rain gauge. The performance of the tipping bucket rain gauge could be monitored from the electrical signal outputted by the tipping bucket rain gauge.

The aim of this study was to investigate the measurement stability of the tipping bucket rain gauge if the water flow rates were varied. The rates were changed between 24 – 100 ml/min three times.

II. THEORY

Rainfall is the width column of rain which is usually measured in millimeter (volume/area) [5]. Mathematical model is shown below [8]:

$$C_h = \frac{V}{A} \tag{1}$$

Where:
C_h= rainfall (mm)
V= volume of rainfall on a certain area (l),
A= section area (m^2).

The section area of the tipping bucket rain gauge is the mouth area of the used tipping bucket rain gauge. The convertion value for 1 mm rainfall with the mouth diameter 20 cm, for example, is equal to the volume of 31.429 ml.

In this study, the section area is 353,571 cm^2. For tipping bucket rain gauge resolution equal to 0.5 mm, the volume of each bucket is around 17.7 ml.

Rainfall is presented in volume per area due to difficulty to measure it directly[9]. Moreover, rain is discrete [10] and dynamic/running measurand [9] and its size might be less than 0.1 mm[11].

Rainfall is a multiplication between rain intensity and rain duration. Although rainfall is different with rain intensity, many rain gauges can be used to measure not only rainfall but also rain intensity[12].

$$I_h = \frac{C_h}{t} = \frac{V}{A.t} \tag{2}$$

$$Q = \frac{V}{t} = I_h.A \tag{3}$$

where:
I_h= rain intensity (mm/h)
C_h= rainfall (mm)
t= rain duration (second, minute, or hour)
A= section area (cm^2)
V= volume (ml)

 978-1-5386-4340-2/18/$31.00 ©2018 IEEE

Q= volume flow rate (ml/min)

From the equation (3), it can be seen that volume flow rate is multiplication between rain intensity and section area. So, we can control volume flow rate of water to a tipping bucket (Figure 1) by making a variation of rain intensity, because the section area is relatively constant.

Every tipping bucket rain gauge has double or twin buckets that are attached each other. There is a lever in the middle of the identical buckets, so they can swing. Therefore, this rain gauge is called tipping bucket rain gauge. This sensor might be used for a rain rose instrument[9].

A Tipping bucket works as a comparison sensor, because it detects the weight of certain water volume which is proportional to the minimum force to tip its pair bucket. The position of these identical buckets can be set by adjusting the height position of their bolt setting. The lower the position of the bolt setting, the heavier the weight of water inside its pair bucket to tip.

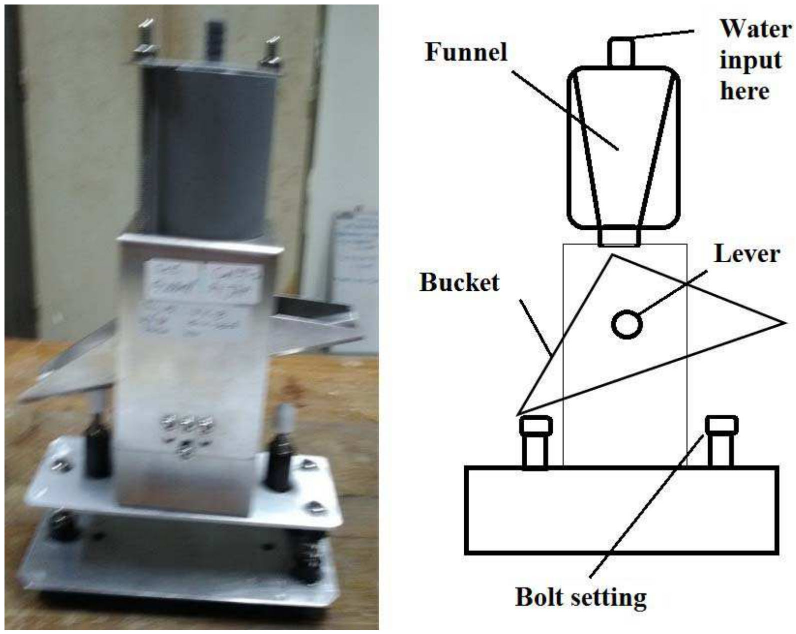

Fig.1. Tipping bucket of rain gauge

Comparing to other rain gauges, a tipping bucket rain gauge has some weaknesses. There is unmeasured water during the movement between a pair of tipping buckets. The error due to the unmeasured water may increase in proportional to the rain intensity.

On the other hand, a tipping bucket rain gauge has its superiority from it easiness to be connected to a processing unit for an automatic measurement. A tipping bucket rain gauge becomes one of WMO international standard devices which spread around the world[13].

A tipping bucket rain gauge is not only an equipment to measure rainfall, but it is also applied to support many measurements, such as rain interception [14], water balance [12] and rain rose [9, 15].

Fig.2. Rain rose apply tipping bucket

III. MEASUREMENT METHOD

The experiments and measurements were conducted between 18th and 20th January 2017 at the temperature of (26.5 ± 0.7) °C and humidity (RH) (54 ± 5) % at a testing laboratory of rain gauges, Research Center for Metrology – Indonesia Institute of Sciences. A measuring tube was used to determine the volume of water. The water flow rate was generated by a pump.

A microcontroller was used as a data acquisition device (Figure 3). The facility of 16 bit counter was applied to read electric signals in the form of pulses from the tested tipping bucket. The servo motor moved the valve of pump. The water flow rate was controlled by the microcontroller using Pulse Width Modulation (PWM) technique [16, 17].

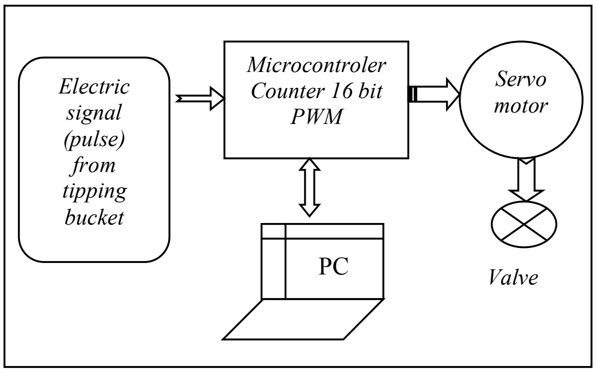

Fig.3. Flow rate control and data processing

Java Application Software (Figure 4) was used for the data acquisition program to avoid a license problem. Java Programming is open source [18]. A serial communication was utilized to facilitate the data recording and processing.

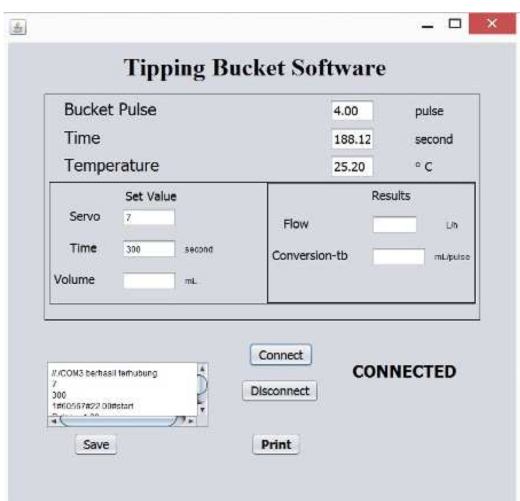

Fig.4. Java application program

The following equations were applied to analysis the data especially in determining the measurement uncertainty due to repeatability [19, 20].

$$\bar{x} = \frac{\sum x_i}{n} \tag{4}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \tag{5}$$

$$u_i = \frac{s}{\sqrt{n}} = \frac{\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}}{\sqrt{n}} \tag{6}$$

$$u_c = \sqrt{\sum c_i^2 . u_i^2} \tag{7}$$

$$U_{ex} = k.u_c \tag{8}$$

Where:

u_i = measurement uncertainty data-i from repeatability

u_c = combined uncertainty

c_i = sensitivity coefficient, 1

s = standard deviation

x_i = data value-i

n = number of data

k = coverage factor due to t-student, 2

U_{ex} = expanded uncertainty.

IV. RESULTS AND DISCUSSION

The pump could generate the water flow rate from 25 to 100 ml/min. The measuring points were 15 points on series. The measurements were carried out 3 times. The results were presented on Figure 5.

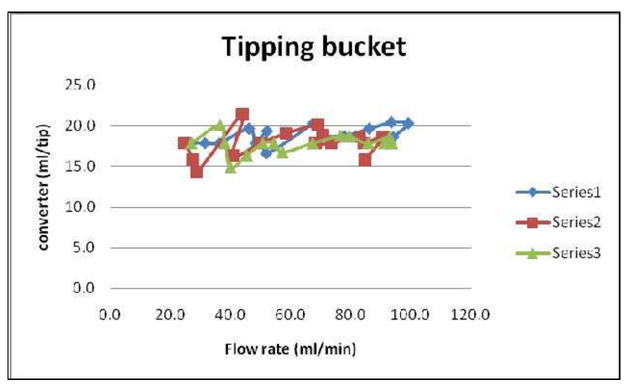

Fig.5. The converter volume vs flow rate

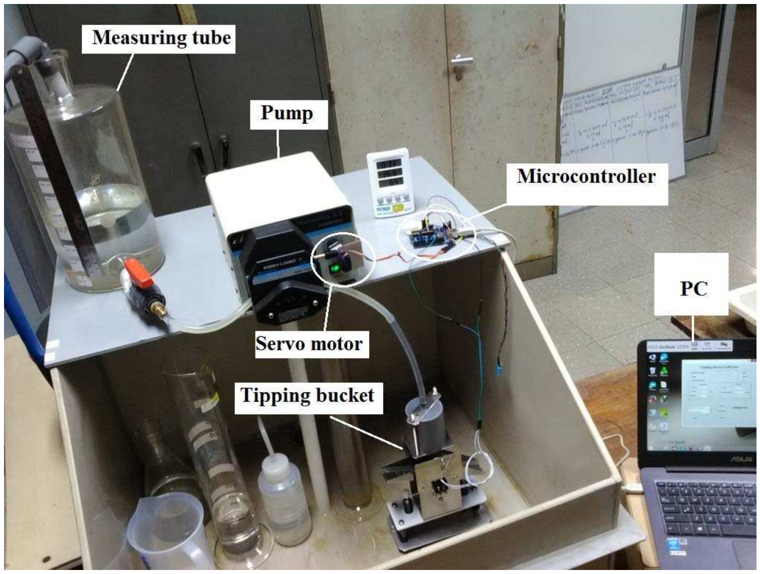

Fig.7. Equipment that was used to test the performance of a tipping bcket rain gauge at the RCM LIPI laboratory

From the Fig 5, it can be seen that converter volumes fluctuated between 14 ml/tip and 21 ml/tip along the flow rate. These fluctuations prove that water flow rate had a significant effect to the performance of tipping bucket. The measurement uncertainty can be used to evaluate how big was this effect (Table 1).

TABLE 1. MEASUREMENT RESULTS WITH UNCERTAINTY

Series	Converter value (ml/tip)	Uncertainty (ml/tip)
1	18.93	0.59
2	17.89	0.92
3	17.80	0.61
Average	18.21	1.25

The series1 was the largest result, namely 18.93 ml/tip. The series1, however, had the smallest uncertainty (0.59 ml/tip or 3%). The series2 had the biggest uncertainty around 0.92 ml/tip or 5% but its converter value was closed to series3.

A further comparison is presented in Fig 6. The graph in Fig 6 shows that the series2 and series3 were closed each other. The converter value in Series1 was the biggest, but its uncertainty was the smallest.

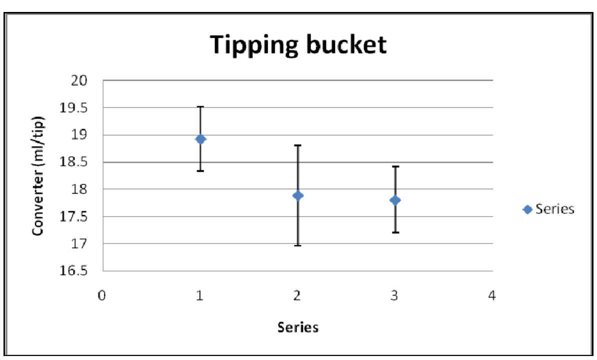

Fig.6. A measurement uncertainty comparison for each series.

The average of all measurements was 18.21 ± 1.25 ml/tip. It means that the measurement uncertainty had a contribution around 7 %. Due to this reason, rain intensity converted from water flow rate had a significant influence to the performance of tipping bucket. It is significant because the error in tipping bucket sensor is usually limited < 5 % and even sometimes < 3 %. The biggest contributor for the error and uncertainty in this testing perhaps due to unmeasured water during tipping bucket rotations and the hysteresis effect of the pump.

From the above explanations, it is clear that a tipping bucket rain gauge should not be used for high rain intensities. If we still would like to utilize it, consider to choose a double layer tipping bucket. A double layer tipping bucket is designed to measure rain for high rain intensities[21].

V. CONCLUSION

The rain intensity made by controlling the water flow rate had significant influence to the performance particularly the measurement repeatability of the tipping bucket rain gauge from measurement uncertainty aspect. The volume in the tipping bucket rain gauge sample was recommended as 17.7 ml. This value might be used to estimate the true value of the tipping bucket volume applied to a rainrose instrument. This method can be implemented to test other tipping bucket rain gauges.

ACKNOWLEDGMENT

The authors would like to express our thank to the management of RCM – LIPI and our research team for supporting this research.

References

[1] Wijonarko, S., Maftukhah, T., Rustandi, D. Damayanti, N.T.E., Permana, D. & Santoso, B., 2016. A method to compare two hyetometer calibrators. PPI KIM, Volume 42, pp. 354-364.

[2] Diani, Fitri., dkk. “Kajian Sistem Informasi Prakiraan Cuaca BMKG Pada BMKG Bandung.” Seminar Nasional Aplikasi Teknologi Informasi (SNATI) 2012. Yogyakarta: 15–16 Juni 2012. ISSN:1907-5022

[3] WMO-No. “Guide to Meteorological Instruments and Methods of Observation”, WMO-No. 8, 2008 edition, updated in 2010, 2012

[4] Yang, D., Goodison, B.E. & Metcalfe, J.R., 1998. Accuracy of NWS 80 Standard Nonrecording Precipitation Gauge: Results and Application of WMO Intercomparison. Journal of Atmospheric and Oceanic Technology, Volume 15, pp. 54-68.

[5] Maftukhah, T., S. Wijonarko, & D. Rustandi 2016. Comparison and correlation among measurement results of observatory, Hellman, and tipping bucket sensors. Instrumentasi scientific publication, vol. 40 (1): 7 -14.

[6] Strangeways, Ian. “Precipitation: Theory, Measurement and Distribution”, Cambridge : Cambridge University Press, 140. ISBN 978-0521172929, 2011

[7] Monjo, R. (2016). "Measure of rainfall time structure using the dimensionless n-index". Climate Research. 67: 71–86. doi:10.3354/cr01359. (pdf) Archived 2017-01-06 at the Wayback Machine, 2016

[8] Herwitz, S.R. & Slye, R.E., “Three-dimensional modeling of canopy tree interception of wind-driven rainfall”. Journal of Hydrology, Volume 168, pp. 205-226, 1995

[9] Wijonarko, S., Maftukhah, T., Rustandi, D. “The compartment area calculation for rainrose measurements”, 2017. Being reviewed.

[10] van Dijk, A.I.J.M. & Bruijnzeel, L.A. “Modeling rainfall interception by vegetation of variable density using an adapted analytical model. Part 2. Model validation for a tropical upland mixed cropping system”. Journal of Hydrology, Volume 247, pp. 239-262, 2001.

[11] Uijlenhoet, R. & Stricker, 1999. Dependence of rainfall on drop size – a comment. Journal of Hydrology, Volume 217, pp. 157-163.

[12] Wijonarko, S. & Maftukhah, T., “Instrumentation system for water balance measurements on Serkuk Subbasin, Kubu Watershed,Belitung”. http://dx.doi.org/10.1063/1.4953930, 2016

[13] WMO-No. 544. “Manual on the global observing system, volume I (annex V to the WMO technical regulations), global aspects, 2003 edition”, 2003

[14] Wijonarko, S. & Maftukhah, T., “Instrumentation development for rainfall interception measurement on a tree using water balance method”. Instrumentasi Scientific Publication, Volume 38 (2), pp. 1-9, 2014

[15] Wijonarko, S., 2007. An examplary embodiment for a method to obtain rainfall information using 48 sensors of a rainrose instrument. Being reviewed.

[16] Prakosa, J.A & Sirenden B.H. 2011. “Designing the Automatic Two-Ways Proportional Solenoid Control Valve through PC on Water Flow Rate Calibrator of Piston Prover OT-400.” Journal Information, Control System and Computer, Vol 5: 57–64. ISSN 1979-8059. Updated in November 2011 http://jurnal.informatika.lipi.go.id/index.php/inkom/issue/view/9/showToc

[17] Prakosa, J.A, Rustandi D., Bernadus H.S, Maftukhah T. “The Automatic Pump Potentiometer Control on the Testing System of Rain Gauge of Tipping Bucket”. Instrumentation Scientific Publication. Vol. 40 No. 2, LIPI Press : Jakarta, 2016

[18] “Stanford Oracle: The Java™ Tutorials.” Updated in Februari 9, 2017. https://docs.oracle.com/javase/tutorial/.

[19] JCGM. “JCGM 100:2008 Evaluation of Measurement Data: Guide to the Expression of Uncertainty in Measurement”. Paris: BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, and OIM, 2008.

[20] Bureau International des Poids et Mesures. “The International System of Units (SI)”. 8th Edition. Paris: Organisation Intergouvernementale de la Convention du Mètre, 2006

[21] Wijonarko, S. The main purpose for the application of double layer tipping bucket sensors in rainfall measurements. Being reviewed.